

Database Standards

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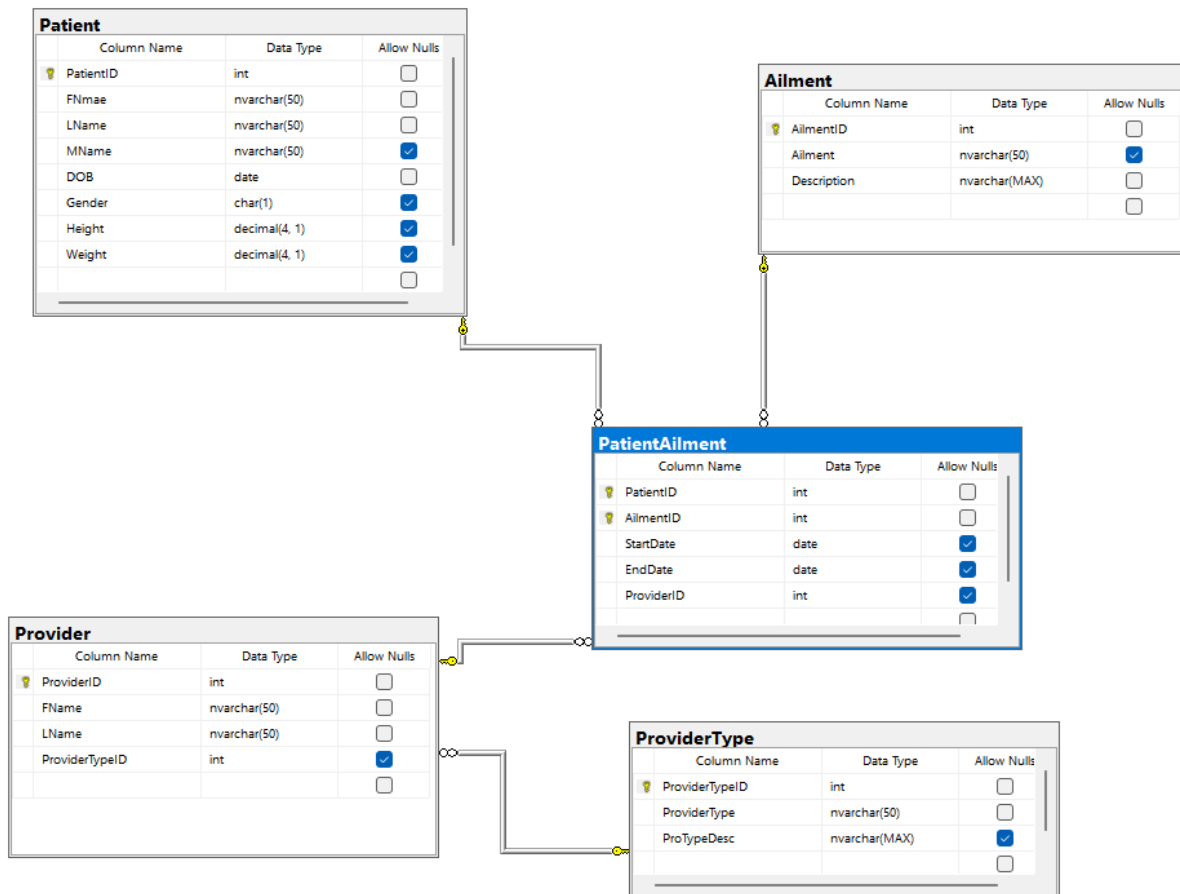
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Database Security ISIN - 325

Program Overview

The Sick Database is a foundational hospital management system designed to replicate on how data is recorded. It consisted of five core tables: Patient, Ailment, PatientAilment, Provider, and ProviderType to record patient demographics, medical conditions, healthcare providers, and their specialization. The Patient Ailment table is a many-to-many relation, allowing patients to have multiple ailments while ailments can affect multiple patients.

Database Design

Sick DB

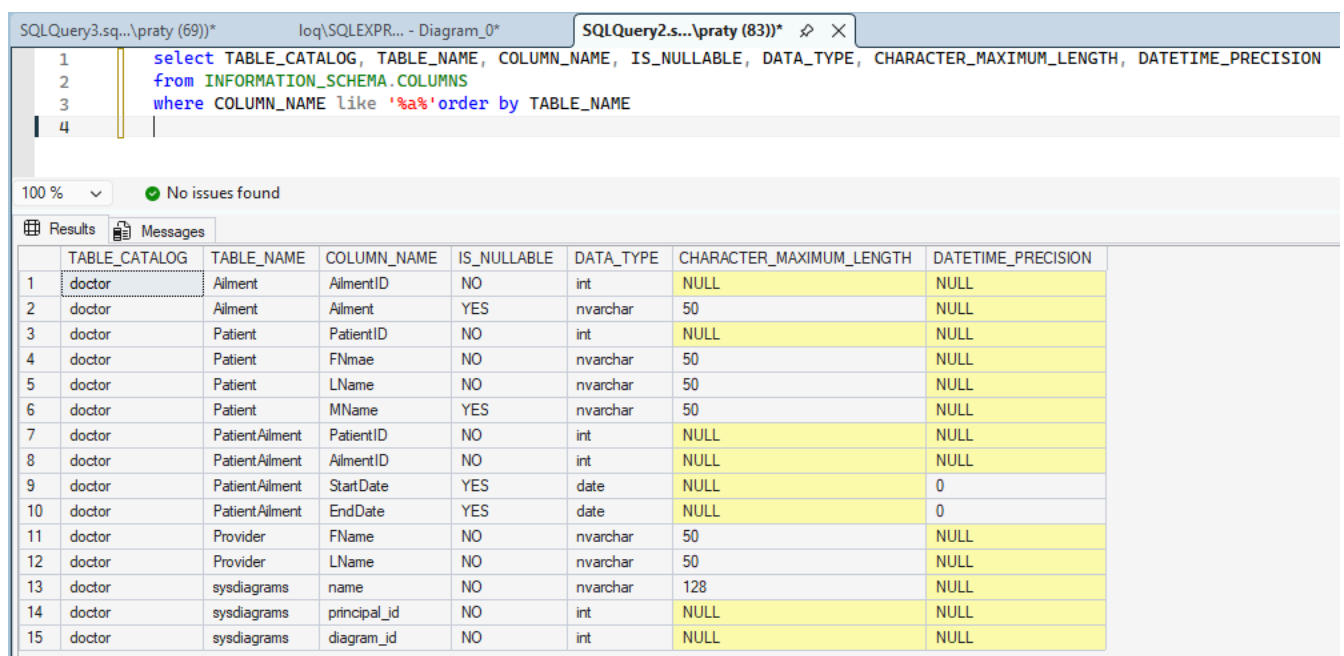


Data Dictionary

A Data Dictionary is a centralized reference document that describes all the data elements, tables, fields, and structures within a database. It serves as a comprehensive guide documenting what each field contains, its data type, format, constraints, and relationships.

A Data Dictionary acts as a communication tool between developers, database administrators, and stakeholders. It ensures everyone understands the database structure, prevents inconsistencies, supports documentation standards, and makes maintenance and future development much easier. For your Sick Database, a Data Dictionary would document fields like PatientID, PatientName, Ailment, and their corresponding properties.

I will be working on with TABLE_CATALOG, TABLE_NAME, COLUMN_NAME, IS_NULLABLE, DATA_TYPE, CHARACTER_MAXIMUM_LENGTH, DATETIME_PRECISION from INFORMATION_SCHEMA.COLUMNS

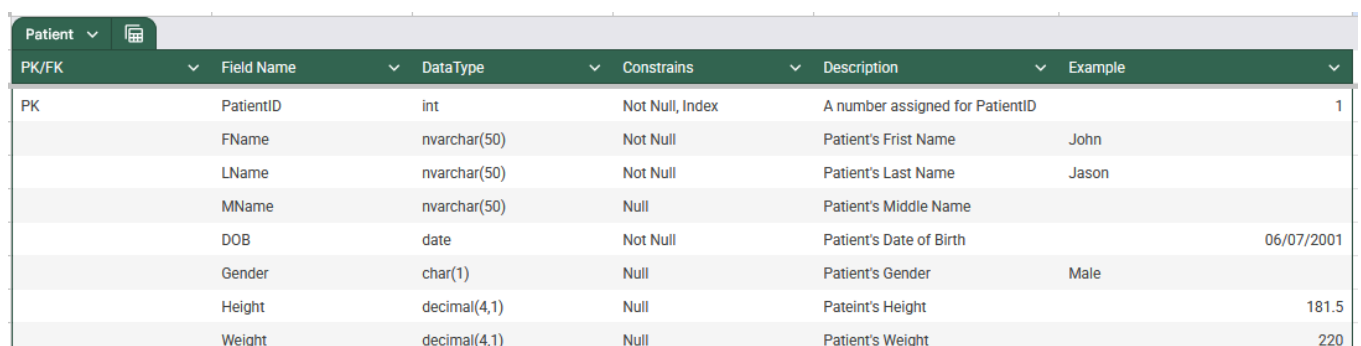


The screenshot shows a SQL Server Enterprise Manager interface. At the top, there are three tabs: 'SQLQuery3.sq... \praty (69))*', 'loq\SQLEXPR... - Diagram_0*', and 'SQLQuery2.s... \praty (83))*'. The active tab shows a SQL query:
1 select TABLE_CATALOG, TABLE_NAME, COLUMN_NAME, IS_NULLABLE, DATA_TYPE, CHARACTER_MAXIMUM_LENGTH, DATETIME_PRECISION
2 from INFORMATION_SCHEMA.COLUMNS
3 where COLUMN_NAME like '%a%' order by TABLE_NAME
4
Below the query, there is a 'Results' tab showing a table with 8 columns: TABLE_CATALOG, TABLE_NAME, COLUMN_NAME, IS_NULLABLE, DATA_TYPE, CHARACTER_MAXIMUM_LENGTH, and DATETIME_PRECISION. The table contains 15 rows of data, with the first column (TABLE_CATALOG) consistently showing 'doctor'.

	TABLE_CATALOG	TABLE_NAME	COLUMN_NAME	IS_NULLABLE	DATA_TYPE	CHARACTER_MAXIMUM_LENGTH	DATETIME_PRECISION
1	doctor	Ailment	AilmentID	NO	int	NULL	NULL
2	doctor	Ailment	Ailment	YES	nvarchar	50	NULL
3	doctor	Patient	PatientID	NO	int	NULL	NULL
4	doctor	Patient	FNmae	NO	nvarchar	50	NULL
5	doctor	Patient	LName	NO	nvarchar	50	NULL
6	doctor	Patient	MName	YES	nvarchar	50	NULL
7	doctor	PatientAilment	PatientID	NO	int	NULL	NULL
8	doctor	PatientAilment	AilmentID	NO	int	NULL	NULL
9	doctor	PatientAilment	StartDate	YES	date	NULL	0
10	doctor	PatientAilment	EndDate	YES	date	NULL	0
11	doctor	Provider	FName	NO	nvarchar	50	NULL
12	doctor	Provider	LName	NO	nvarchar	50	NULL
13	doctor	sysdiagrams	name	NO	nvarchar	128	NULL
14	doctor	sysdiagrams	principal_id	NO	int	NULL	NULL
15	doctor	sysdiagrams	diagram_id	NO	int	NULL	NULL

Patient Table

This table contains detailed demographic information about the Patient.



The screenshot shows a table with 7 columns: PK/FK, Field Name, DataType, Constrains, Description, and Example. The table contains 8 rows of data, representing the Patient table. The first row is the primary key (PK) for PatientID. The other rows represent the fields: FName, LName, MName, DOB, Gender, Height, and Weight, each with their respective data types, constraints, descriptions, and examples.

PK/FK	Field Name	DataType	Constrains	Description	Example
PK	PatientID	int	Not Null, Index	A number assigned for PatientID	1
	FName	nvarchar(50)	Not Null	Patient's Frist Name	John
	LName	nvarchar(50)	Not Null	Patient's Last Name	Jason
	MName	nvarchar(50)	Null	Patient's Middle Name	
	DOB	date	Not Null	Patient's Date of Birth	06/07/2001
	Gender	char(1)	Null	Patient's Gender	Male
	Height	decimal(4,1)	Null	Pateint's Height	181.5
	Weight	decimal(4,1)	Null	Patient's Weight	220

Ailment Table

This table contains information on the patient's sickness, where they can describe their symptoms in Description (Field Type). The Ailment's data type is null because the patient may not know what their illness is called.

Ailment						
PK/FK	Field Name	DataType	Constrains	Description	Example	
PK	AilmentID	int	Not Null, Index	A number assigned for Ailem		1
	Ailment	nvarchar(50)	Null	A number assigned to Ailmer	Hypertension	
	Description	nvarchar(max)	Not Null	Description of sickness	Severe headaches, Dizziness, Nosebleeds	

PatientAilment Table

This table has a many-to-many relation with the Patient Table and the Ailment Table, cause same Patient can have many Ailments and vice versa.

PatientAilment						
PK/FK	Field Name	DataType	Constrains	Description	Example	
PK	PatientID	int	Not Null, Index	A number assigned for PatientID		1
PK	AilmentID	int	Not Null, Index	A number assigned for AilmentID		6
	StartDate	date	Null	When did the symptoms started showing	03/22/2026	
	EndDate	date	Null	When did the sickness stopped		04/02/2026
FK	ProviderID	int	Null	A number assigned for Provider		3

Provider Table

This table contains the Providers' (Doctors) information along with their specialization.

Provider						
PK/FK	Field Name	DataType	Constrains	Description	Example	
PK	ProviderID	int	Not Null, Index	A number assigned for Provider		1
	FName	nvarchar(50)	Not Null	First Name of Provider	Eva	
	LName	nvarchar(50)	Not Null	Last Name of Provider	Green	
FK	ProviderTypeID	int	Null	A number provided to Provider to know their specialization		4

ProviderType Table

This table stores information on provider descriptions of treatment methods or what medication was used.

ProviderType						
PK/FK	Field Name	DataType	Constrains	Description	Example	
PK	ProviderTypeID	int	Not Null, Index	A number to represent Providers' specialization		1
	ProviderType	nvarchar(50)	Not Null	Providers' specialization	Cardiologists	
	ProTypeDesc	nvarchar(max)	Null	Description provided by Provider (Doctor)	Recommended Medication	

Standards Document

There is a certain level of quality that must be met when creating a database. Each business or organization will form its own standards, but most follow the standards that were given below.

- Use of camel case for naming tables and fields. (FName, LName)
- Use of Windows logins and not SQL Server Logins (Windows domain)
- Use PRIMARY KEYS and FOREIGN KEYS to enforce relationships between tables.
- Establish regular automated backups and documented disaster recovery procedures.
- Use appropriate data types for each field (int for IDs, varchar for names, date for dates, decimal for currency).

Summary

Working on the Sick Database provided me the basics of databases and how the data is organized. Though this is only a basic replica of a Hospital Database system, it showed me the core concepts, including entity relationships, data normalization, and referential integrity.

After working on this project, I realized that I can make it more complex with some other Field Types; Contact Information of Providers, Treatment Records, multiple many-to-many relations. But since class is about Database Security, I would like to learn about SQL-injections, and some other data corruption, so that we can fix them.